Ablation performance

Six variants evaluated on the selected graph.

(a) Adaptability (Ada)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	**0.790 ± 0.001**	0.405 ± 0.041	0.699 ± 0.000	0.723 ± 0.010	0.773 ± 0.004	0.683 ± 0.025
10	0.743 ± 0.005	0.425 ± 0.033	0.699 ± 0.000	0.699 ± 0.018	**0.745 ± 0.003**	0.674 ± 0.016
15	0.716 ± 0.020	0.426 ± 0.034	0.699 ± 0.000	0.681 ± 0.021	**0.721 ± 0.012**	0.664 ± 0.011
20	**0.712 ± 0.017**	0.410 ± 0.015	0.700 ± 0.002	0.669 ± 0.025	0.711 ± 0.009	0.652 ± 0.006
25	0.692 ± 0.004	0.410 ± 0.015	**0.703 ± 0.003**	0.656 ± 0.025	0.692 ± 0.006	0.642 ± 0.004
30	0.673 ± 0.006	0.414 ± 0.013	**0.708 ± 0.003**	0.645 ± 0.023	0.675 ± 0.007	0.631 ± 0.004
35	0.662 ± 0.010	0.409 ± 0.017	**0.711 ± 0.003**	0.634 ± 0.023	0.665 ± 0.004	0.623 ± 0.004
40	0.655 ± 0.006	0.400 ± 0.019	**0.715 ± 0.001**	0.624 ± 0.024	0.658 ± 0.003	0.615 ± 0.006
45	0.649 ± 0.004	0.392 ± 0.021	**0.715 ± 0.001**	0.613 ± 0.023	0.653 ± 0.002	0.607 ± 0.008
50	0.644 ± 0.007	0.386 ± 0.023	**0.714 ± 0.002**	0.603 ± 0.022	0.649 ± 0.003	0.599 ± 0.008

(b) Novelty (NOV)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	**0.993 ± 0.001**	0.975 ± 0.002	0.991 ± 0.000	0.990 ± 0.001	0.991 ± 0.002	0.986 ± 0.006
10	0.988 ± 0.001	0.975 ± 0.001	**0.991 ± 0.000**	0.989 ± 0.003	0.986 ± 0.001	0.985 ± 0.004
15	0.985 ± 0.001	0.975 ± 0.001	**0.991 ± 0.000**	0.988 ± 0.003	0.984 ± 0.001	0.985 ± 0.004
20	0.984 ± 0.000	0.974 ± 0.001	**0.991 ± 0.000**	0.987 ± 0.004	0.984 ± 0.001	0.985 ± 0.004
25	0.985 ± 0.000	0.974 ± 0.001	**0.991 ± 0.000**	0.987 ± 0.003	0.984 ± 0.000	0.986 ± 0.003
30	0.985 ± 0.001	0.974 ± 0.001	**0.991 ± 0.000**	0.987 ± 0.003	0.984 ± 0.000	0.985 ± 0.003
35	0.985 ± 0.001	0.974 ± 0.001	**0.991 ± 0.000**	0.987 ± 0.004	0.984 ± 0.000	0.986 ± 0.003
40	0.985 ± 0.001	0.973 ± 0.002	**0.991 ± 0.000**	0.987 ± 0.003	0.985 ± 0.000	0.986 ± 0.003
45	0.985 ± 0.000	0.973 ± 0.001	**0.991 ± 0.000**	0.987 ± 0.003	0.985 ± 0.000	0.986 ± 0.002
50	0.985 ± 0.000	0.973 ± 0.001	**0.990 ± 0.001**	0.987 ± 0.002	0.985 ± 0.000	0.986 ± 0.002

(theta_m, theta_f, rec) = (0.8, 0.8, 50); zero exercise-forgetting edges removed.

Comparison performance

Eight baselines and 2CKG4ER evaluated on the selected graph.

(a) Adaptability (Ada)

N	EB-CF	SB-CF	CBF	TransE	TransE-adv	RotatE	DistMult	ComplEx	2CKG4ER
5	0.222	0.450	0.274	0.730 ± 0.005	0.725 ± 0.005	**0.791 ± 0.004**	0.509 ± 0.030	0.378 ± 0.016	0.790 ± 0.001
10	0.220	0.515	0.269	0.701 ± 0.005	0.711 ± 0.008	**0.760 ± 0.002**	0.479 ± 0.021	0.357 ± 0.010	0.743 ± 0.005
15	0.219	0.498	0.269	0.672 ± 0.005	0.689 ± 0.004	**0.725 ± 0.004**	0.450 ± 0.016	0.343 ± 0.004	0.716 ± 0.020
20	0.217	0.468	0.271	0.653 ± 0.006	0.676 ± 0.006	0.697 ± 0.003	0.432 ± 0.013	0.332 ± 0.003	**0.712 ± 0.017**
25	0.218	0.441	0.272	0.637 ± 0.008	0.668 ± 0.008	0.677 ± 0.006	0.416 ± 0.012	0.326 ± 0.001	**0.692 ± 0.004**
30	0.217	0.418	0.272	0.624 ± 0.009	0.659 ± 0.008	0.665 ± 0.006	0.405 ± 0.012	0.321 ± 0.000	**0.673 ± 0.006**
35	0.217	0.398	0.273	0.613 ± 0.008	0.654 ± 0.009	0.655 ± 0.007	0.395 ± 0.011	0.316 ± 0.002	**0.662 ± 0.010**
40	0.216	0.381	0.275	0.604 ± 0.009	0.649 ± 0.010	0.644 ± 0.003	0.387 ± 0.012	0.313 ± 0.004	**0.655 ± 0.006**
45	0.216	0.366	0.275	0.594 ± 0.009	0.642 ± 0.010	0.633 ± 0.004	0.380 ± 0.011	0.309 ± 0.004	**0.649 ± 0.004**
50	0.216	0.355	0.274	0.586 ± 0.009	0.633 ± 0.010	0.621 ± 0.005	0.375 ± 0.009	0.306 ± 0.004	**0.644 ± 0.007**

(b) Novelty (NOV)

N	EB-CF	SB-CF	CBF	TransE	TransE-adv	RotatE	DistMult	ComplEx	2CKG4ER
5	0.957	0.975	0.993	0.991 ± 0.001	**0.996 ± 0.001**	0.993 ± 0.000	0.988 ± 0.001	0.986 ± 0.003	0.993 ± 0.001
10	0.961	0.979	0.994	0.992 ± 0.000	**0.994 ± 0.000**	0.990 ± 0.000	0.986 ± 0.001	0.987 ± 0.003	0.988 ± 0.001
15	0.964	0.982	**0.995**	0.994 ± 0.000	0.993 ± 0.001	0.989 ± 0.000	0.986 ± 0.002	0.986 ± 0.002	0.985 ± 0.001
20	0.964	0.981	0.994	**0.995 ± 0.000**	0.994 ± 0.000	0.989 ± 0.000	0.986 ± 0.003	0.986 ± 0.002	0.984 ± 0.000
25	0.965	0.981	0.991	**0.994 ± 0.000**	0.993 ± 0.001	0.989 ± 0.000	0.986 ± 0.003	0.986 ± 0.002	0.985 ± 0.000
30	0.966	0.981	0.990	**0.993 ± 0.000**	0.993 ± 0.001	0.990 ± 0.000	0.987 ± 0.002	0.986 ± 0.001	0.985 ± 0.001
35	0.965	0.980	0.990	**0.993 ± 0.000**	0.992 ± 0.000	0.990 ± 0.000	0.987 ± 0.002	0.986 ± 0.001	0.985 ± 0.001
40	0.965	0.979	0.990	**0.993 ± 0.000**	0.992 ± 0.001	0.990 ± 0.000	0.987 ± 0.001	0.985 ± 0.000	0.985 ± 0.001
45	0.965	0.980	0.990	**0.993 ± 0.000**	0.991 ± 0.001	0.990 ± 0.000	0.987 ± 0.001	0.985 ± 0.000	0.985 ± 0.000
50	0.964	0.980	0.990	**0.993 ± 0.000**	0.991 ± 0.000	0.990 ± 0.000	0.987 ± 0.001	0.985 ± 0.000	0.985 ± 0.000

(theta_m, theta_f, rec) = (0.8, 0.8, 50); zero exercise-forgetting edges removed.